

Exercise 1A — Full Solutions

Questions 1–10 (complete)

Roots of polynomial equations: sum and product of roots

Colour key: *green* = verification check | *blue* = alternative method.

Question 1

Each quadratic has roots α, β . Find $\alpha + \beta$ and $\alpha\beta$, i.e. read off $-\frac{b}{a}$ and $\frac{c}{a}$.

(a) $x^2 + 5x + 9 = 0$ ($a = 1, b = 5, c = 9$): $\alpha + \beta = -5$, $\alpha\beta = 9$.

(b) $x^2 - 4x + 8 = 0$ ($a = 1, b = -4, c = 8$): $\alpha + \beta = 4$, $\alpha\beta = 8$.

(c) $2x^2 + 3x - 7 = 0$ ($a = 2, b = 3, c = -7$): $\alpha + \beta = -\frac{3}{2}$, $\alpha\beta = -\frac{7}{2}$.

Check. Signs behave as expected: in (a) both sum and product positive (the discriminant $25 - 36 < 0$ gives a complex conjugate pair); in (c) the negative product reflects real roots of opposite sign, since $\alpha\beta = c/a < 0$ ✓.

Question 2

Given that $3x^2 + 4x + 12 = 0$ has roots α, β , find (a) $\alpha + \beta$ and $\alpha\beta$; (b) $\alpha^2 + \beta^2$.

(a) For $ax^2 + bx + c = 0$ we use $\alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$. Here $a = 3$, $b = 4$, $c = 12$:

$$\alpha + \beta = -\frac{4}{3}, \quad \alpha\beta = \frac{12}{3} = 4.$$

(b) Use the identity $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$:

$$\alpha^2 + \beta^2 = \left(-\frac{4}{3}\right)^2 - 2(4) = \frac{16}{9} - 8 = \frac{16 - 72}{9} = -\frac{56}{9}.$$

Check. The discriminant is $b^2 - 4ac = 16 - 144 = -128 < 0$, so the roots are a complex conjugate pair. For such roots $\alpha^2 + \beta^2$ can be negative, which is consistent with $-\frac{56}{9} < 0$.

Question 3

$x^2 - (2 + p)x + (7 + p) = 0$ has roots that differ by 1. Find p , given $p > 0$.

Let the roots be α, β with $\alpha + \beta = 2 + p$ and $\alpha\beta = 7 + p$. “Differ by 1” means $(\alpha - \beta)^2 = 1$. Using $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$:

$$(2 + p)^2 - 4(7 + p) = 1 \implies 4 + 4p + p^2 - 28 - 4p = 1 \implies p^2 - 24 = 1 \implies p^2 = 25.$$

So $p = \pm 5$, and since $p > 0$, $\boxed{p = 5}$.

Check. With $p = 5$ the equation is $x^2 - 7x + 12 = 0$, i.e. $(x - 3)(x - 4) = 0$. The roots 3 and 4 differ by 1 ✓.

Question 4

If $a + b = -3$ and $a^2 + b^2 = 7$, find the value of ab and hence write down a quadratic equation with roots a and b .

Rearrange $a^2 + b^2 = (a + b)^2 - 2ab$ to isolate ab :

$$7 = (-3)^2 - 2ab \implies 7 = 9 - 2ab \implies 2ab = 2 \implies ab = 1.$$

A quadratic with roots a and b is $x^2 - (a + b)x + ab = 0$:

$$x^2 - (-3)x + 1 = 0 \implies \boxed{x^2 + 3x + 1 = 0}.$$

Check. For $x^2 + 3x + 1 = 0$: sum of roots $= -3$ ✓ and product $= 1$ ✓. Then $a^2 + b^2 = (a + b)^2 - 2ab = 9 - 2 = 7$ ✓, as required.

Question 5

$x^2 + bx + c = 0$ has roots α, β , so $\alpha + \beta = -b$ and $\alpha\beta = c$. Prove: **(a)** if $\alpha = 3\beta$ then $b^2 = \frac{16}{3}c$; **(b)** if $\alpha = \beta - 2$ then $b^2 = 4(c + 1)$.

(a) With $\alpha = 3\beta$: from the sum, $\alpha + \beta = 4\beta = -b$, so $\beta = -\frac{b}{4}$; from the product, $\alpha\beta = 3\beta^2 = c$, so $\beta^2 = \frac{c}{3}$. Equating β^2 :

$$\left(-\frac{b}{4}\right)^2 = \frac{c}{3} \implies \frac{b^2}{16} = \frac{c}{3} \implies b^2 = \frac{16}{3}c. \quad \blacksquare$$

(b) With $\alpha = \beta - 2$ we have $\alpha - \beta = -2$, so $(\alpha - \beta)^2 = 4$. Using $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = b^2 - 4c$:

$$b^2 - 4c = 4 \implies b^2 = 4c + 4 = 4(c + 1). \quad \blacksquare$$

Check. (a) $\beta = 2, \alpha = 6$ give $b = -8, c = 12$: $b^2 = 64 = \frac{16}{3}(12)$ ✓. (b) $\beta = 5, \alpha = 3$ give $b = -8, c = 15$: $b^2 = 64 = 4(15 + 1)$ ✓.

Question 6

The quadratic $px^2 + qx - 16 = 0$ has roots α, β with $\alpha + \beta = -\frac{1}{2}$ and $\alpha\beta = -8$. Find p and q .

Here $a = p$, $b = q$, $c = -16$, so

$$\alpha\beta = \frac{c}{a} = \frac{-16}{p}, \quad \alpha + \beta = -\frac{b}{a} = -\frac{q}{p}.$$

Use the product first (it contains only p):

$$-8 = \frac{-16}{p} \implies p = \frac{-16}{-8} = 2.$$

Now use the sum with $p = 2$:

$$-\frac{1}{2} = -\frac{q}{2} \implies q = 1.$$

Hence $\boxed{p = 2, q = 1}$.

Check. The equation is $2x^2 + x - 16 = 0$. Then $\alpha + \beta = -\frac{1}{2}$ ✓ and $\alpha\beta = \frac{-16}{2} = -8$ ✓.

Question 7

$x^2 + 2x - 6 = 0$ has roots α, β . Find $(\alpha - \beta)^2$ and $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$.

Here $\alpha + \beta = -2$ and $\alpha\beta = -6$.

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = (-2)^2 - 4(-6) = 4 + 24 = 28.$$

For the second, combine over a common denominator and use $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$:

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{(\alpha\beta)^2} = \frac{(-2)^2 - 2(-6)}{(-6)^2} = \frac{4 + 12}{36} = \frac{16}{36} = \frac{4}{9}.$$

Check. Numerically $(\alpha - \beta)^2 = 28.000$ and $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = 0.444 = \frac{4}{9} \checkmark$.

Question 8

A quadratic equation has roots α, β with $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{1}{2}$ and $\alpha^2 + \beta^2 = 12$. Find two possible quadratic equations.

Let $s = \alpha + \beta$ and $p = \alpha\beta$. Combine the first condition over a common denominator:

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{s}{p} = \frac{1}{2} \implies p = 2s.$$

Substitute into $\alpha^2 + \beta^2 = s^2 - 2p = 12$:

$$s^2 - 2(2s) = 12 \implies s^2 - 4s - 12 = 0 \implies (s - 6)(s + 2) = 0,$$

so $s = 6$ or $s = -2$, giving $p = 2s = 12$ or $p = -4$ respectively.

Using $x^2 - sx + p = 0$, the two possible equations are

$$\boxed{x^2 - 6x + 12 = 0} \quad \text{and} \quad \boxed{x^2 + 2x - 4 = 0}.$$

Check. For $x^2 - 6x + 12 = 0$: $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{6}{12} = \frac{1}{2} \checkmark$ and $\alpha^2 + \beta^2 = 36 - 24 = 12 \checkmark$.

For $x^2 + 2x - 4 = 0$: $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{-2}{-4} = \frac{1}{2} \checkmark$ and $\alpha^2 + \beta^2 = 4 - 2(-4) = 12 \checkmark$.

Question 9

The quadratic $3x^2 + 2x - 4 = 0$ has roots α, β . Find the values of S_1 , S_2 and S_{-1} , where $S_n = \alpha^n + \beta^n$.

Here $a = 3$, $b = 2$, $c = -4$, so

$$\alpha + \beta = -\frac{2}{3}, \quad \alpha\beta = -\frac{4}{3}.$$

S_1 : by definition $S_1 = \alpha + \beta = -\frac{2}{3}$.

S_2 : use $S_2 = \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$:

$$S_2 = \left(-\frac{2}{3}\right)^2 - 2\left(-\frac{4}{3}\right) = \frac{4}{9} + \frac{8}{3} = \frac{4 + 24}{9} = \frac{28}{9}.$$

S_{-1} : use $S_{-1} = \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$:

$$S_{-1} = \frac{-\frac{2}{3}}{-\frac{4}{3}} = \frac{1}{2}.$$

Alternative — S_n recurrence. Multiplying each root's equation $3r^2 + 2r - 4 = 0$ by r^n and summing over both roots gives, for all integers n ,

$$3S_{n+2} + 2S_{n+1} - 4S_n = 0, \quad \text{with } S_0 = 2.$$

Then $n = 0$: $3S_2 + 2S_1 - 4S_0 = 0 \Rightarrow 3S_2 + 2(-\frac{2}{3}) - 8 = 0 \Rightarrow S_2 = \frac{28}{9}$.

And $n = -1$: $3S_1 + 2S_0 - 4S_{-1} = 0 \Rightarrow -2 + 4 - 4S_{-1} = 0 \Rightarrow S_{-1} = \frac{1}{2}$.

Check. Solving directly, $\alpha \approx -1.5352$, $\beta \approx 0.8685$. Then $S_1 \approx -0.667 = -\frac{2}{3} \checkmark$, $S_2 \approx 3.111 = \frac{28}{9} \checkmark$, $S_{-1} \approx 0.500 = \frac{1}{2} \checkmark$.

Question 10

The quadratic $4x^2 - x + 6 = 0$ has roots α, β . **(a)** Find $\alpha^2 + \beta^2$. **(b)** Without solving, state what this tells you about the roots.

(a) Here $a = 4$, $b = -1$, $c = 6$, so

$$\alpha + \beta = -\frac{-1}{4} = \frac{1}{4}, \quad \alpha\beta = \frac{6}{4} = \frac{3}{2}.$$

Then

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(\frac{1}{4}\right)^2 - 2\left(\frac{3}{2}\right) = \frac{1}{16} - 3 = -\frac{47}{16}.$$

(b) If α and β were both real, then $\alpha^2 + \beta^2$ would be a sum of two squares and so could not be negative. Since $\alpha^2 + \beta^2 = -\frac{47}{16} < 0$, the roots **cannot be real** — they are a (complex) conjugate pair.

Check. The discriminant is $b^2 - 4ac = (-1)^2 - 4(4)(6) = 1 - 96 = -95 < 0$, confirming the roots are non-real, in agreement with part (b).